

IDEALS ATTAINING EQUALITY IN THE DEMAILLY–PHAM INEQUALITY

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ABSTRACT. Let $I \subset \mathbb{C}[[x_1, \dots, x_n]]$ have height at least l . We prove that $\text{lct}(I) = E_l(I)$ if and only if the integral closure of I is the integral closure of $(y_1^{d_1}, \dots, y_l^{d_l})$ in a regular system of parameters, as conjectured by Baily. The primary case follows from a simultaneous induction for ideals and divisorial valuations. The passage to arbitrary codimension uses transverse primary truncations and compatible finite-order coordinate changes. The diagonal exponents are the successive ratios of the extended mixed multiplicities.

1. INTRODUCTION

Let $\widehat{R} = \mathbb{C}[[x_1, \dots, x_n]]$, with maximal ideal $\widehat{\mathfrak{m}}$. For an ideal I of height at least l , set

$$\sigma_j(I) = \sup_{a \geq 1} e((I + \widehat{\mathfrak{m}}^a)^{[j]}, \widehat{\mathfrak{m}}^{[n-j]}) \quad (1 \leq j \leq l),$$

$$\sigma_0(I) = 1, \quad E_l(I) = \sum_{j=0}^{l-1} \frac{\sigma_j(I)}{\sigma_{j+1}(I)}.$$

These extended mixed multiplicities are finite. Equivalently, $\sigma_j(I)$ is the multiplicity of I on a sufficiently general j -dimensional linear section. The Demailly–Pham inequality [5], applied on a general l -dimensional section, gives

$$(1.1) \quad E_l(I) \leq \text{lct}(I).$$

The equality case concerns the integral closure of I , since both sides are invariant under integral closure. Bivià-Ausina [2] characterized primary diagonal ideals using equality and a condition on the containing monomial ideal. Baily [1, Theorem 5.1] classified homogeneous equality ideals and formulated the corresponding local classification in [1, Conjecture 6.2]. Our main result is this local classification in every codimension, obtained by algebraic methods.

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Theorem 1.1 (Equality ideals in arbitrary codimension). *Let $0 \neq I \subsetneq \widehat{R} = \mathbb{C}[[x_1, \dots, x_n]]$ and $1 \leq l \leq \text{ht } I$. Then $\text{lct}(I) = E_l(I)$ if and only if there exist a regular system of parameters $y_1, \dots, y_n$ of $\widehat{R}$ and positive integers $d_1 \leq \dots \leq d_l$ such that*

$$(1.2) \quad \overline{I} = \overline{(y_1^{d_1}, \dots, y_l^{d_l})}.$$

In this case $\text{ht } I = l$ and

$$(1.3) \quad d_j = \frac{\sigma_j(I)}{\sigma_{j-1}(I)} \quad (1 \leq j \leq l), \quad \text{lct}(I) = \sum_{j=1}^l \frac{1}{d_j}.$$

The parameters belong to the original complete local ring. Thus the conclusion is a classification up to a formal change of coordinates and integral closure. In particular, equality forces the reduced support of I to be smooth. Section 5 proves a separate reduction theorem showing that the primary classification in dimension n implies the classification for every $l \leq n$ in that same dimension.

1.1. Method of proof and organization. We first prove the primary classification by a simultaneous induction for ideals and divisorial valuations. In dimension $n \geq 2$, put $R = \mathbb{C}[x_1, \dots, x_n]_{(x_1, \dots, x_n)}$ and $\mathfrak{m} = (x_1, \dots, x_n)R$. For a valuation v centered at $\mathfrak{m}$ with finite log discrepancy, write $\mathfrak{a}_m(v) = \{f : v(f) \geq m\}$ and let $\text{vol}(v)$ be the asymptotic colength of its valuation ideals, normalized by $m^n/n!$. Two comparison statements enter this induction. First, the DP function is nonincreasing on coordinatewise increasing segments of the positive log-convex cone; equality for two comparable vectors forces equality of all coordinates. Second, for any nonzero $f \in \mathfrak{m}$,

$$(1.4) \quad \lim_{m \rightarrow \infty} \frac{e(\mathfrak{a}_m(v); R/(f))}{m^{n-1}} = v(f) \text{vol}(v).$$

The latter identity follows from the exact colon formula and Cutkosky's multiplicity theorem [4]. These inputs are established in Sections 2 and 3.

Section 4 proves Proposition 4.1 by simultaneous induction. An equality ideal determines a unique normalized divisorial valuation computing its threshold. Blum's results [3] supply a generated Veronese of its valuation-ideal algebra and a plt extraction. A general hyperplane preserves the equality, and [12, Lemma 2.17] controls its Rees degeneration. Normality of the quotient Rees algebra and reducedness of its central fiber allow us to identify the entire quotient filtration and lift monomial parameters from dimension $n - 1$ to dimension n .

Section 5 passes from primary ideals to arbitrary codimension. If B is generated by $n - l$ sufficiently general linear forms, comparison of mixed multiplicities gives

$$E_n(I + B^N) \geq E_l(I) + \frac{n - l}{N} \quad (N \gg 0).$$

The upper bound for $\text{lct}(I + B^N)$ obtained from the sum of ideals is the same when I is an equality ideal. Thus every sufficiently high transverse truncation is a primary equality ideal. Its diagonal exponents are fixed by the mixed multiplicities, except for the $n - l$ exponents equal to N . Integral-closure continuity then produces coordinate equivalences to every finite order. We pass to compatible jets by the stabilization of algebraic subgroup cosets; this step does not assume that the coordinate changes initially supplied by the primary theorem are compatible.

2. MIXED MULTIPLICITIES, THRESHOLDS, AND VALUATIONS

For the valuative arguments put $R = \mathbb{C}[x_1, \dots, x_n]_{(x_1, \dots, x_n)}$, with maximal ideal $\mathfrak{m}$, and let o be its closed point. Unless specified otherwise, $n \geq 2$ in these arguments. For a primary ideal K , write

$$e_j(K) = e(K^{[j]}, \mathfrak{m}^{[n-j]}), \quad e_0(K) = 1, \quad \text{DP}(K) = E_n(K).$$

For a real valuation v centered at o , let $A(v)$ denote its log discrepancy and put

$$\mathfrak{a}_t(v) = \{f \in R : v(f) \geq t\}, \quad \text{vol}(v) = \lim_{m \rightarrow \infty} \frac{n! \ell(R/\mathfrak{a}_m(v))}{m^n}.$$

When $A(v) < \infty$, the asymptotic mixed multiplicities

$$(2.1) \quad b_0(v) = 1, \quad b_j(v) = \lim_{m \rightarrow \infty} \frac{e_j(\mathfrak{a}_m(v))}{m^j} \quad (1 \leq j \leq n)$$

exist and are positive, with $b_n(v) = \text{vol}(v)$. Set

$$\text{DP}(v) = \sum_{j=0}^{n-1} \frac{b_j(v)}{b_{j+1}(v)}.$$

The finite-level inequality implies

$$(2.2) \quad \text{DP}(v) \leq \text{lct}(\mathfrak{a}_\bullet(v)) \leq A(v).$$

The following subsections give the conventions and proofs needed for these asymptotic statements.

A valuation is monomial in regular parameters $z_1, \dots, z_n$ if the value of a nonzero series $\sum_{\beta} c_{\beta} z^{\beta}$ is $\min_{c_{\beta} \neq 0} \sum_i \beta_i v(z_i)$. Here the parameters lie in the completed local ring.

All valuations considered below are finite on nonzero elements. For a nonzero ideal I , $v(I)$ is the minimum of the values of any finite generating set. We write $e(I; S)$ for the Hilbert–Samuel multiplicity of an ideal of finite colength in a local ring S and $\bar{I}$ for integral closure.

2.1. Ideals of positive-dimensional support. We recall the facts needed for the transverse truncation argument. They apply to both R and $\widehat{R}$; in this subsection write $(S, \mathfrak{n})$ for either ring. For ideals $I_1, \dots, I_n$ set

$$\sigma(I_1, \dots, I_n) = \sup_{a \geq 1} e(I_1 + \mathfrak{n}^a, \dots, I_n + \mathfrak{n}^a).$$

It is finite if one can choose $g_i \in I_i$ whose common ideal is $\mathfrak{n}$ -primary. In that case it equals the multiplicity of $(g_1, \dots, g_n)$ for general choices, and is no larger than that multiplicity for any such parameter tuple. It decreases when any of its ideal arguments is enlarged. See [7, Chapter 17] and [1, Proposition 2.37]. In particular,

$$\sigma_j(I) = \sigma(I^{[j]}, \mathfrak{n}^{[n-j]}), \quad \sigma_j(I)^2 \leq \sigma_{j-1}(I)\sigma_{j+1}(I)$$

whenever these quantities are finite. Finiteness holds exactly for $j \leq \text{ht } I$.

For $r = n - l$, choose general linear forms $z_1, \dots, z_r$ and put $S_L = S/(z_1, \dots, z_r)$. The image IS_L has finite colength and multiplicity $\sigma_l(I)$. Choose l general elements of I whose images generate a reduction of IS_L . Since S_L is Cohen–Macaulay, their parameter-ideal multiplicity is their colength. Thus there are $f_1, \dots, f_l \in I$ with

$$(2.3) \quad \ell(S/(f_1, \dots, f_l, z_1, \dots, z_r)) = \sigma_l(I).$$

This tuple is a regular sequence. Replacing any of the z_i by its N -th power multiplies its colength by N .

We use the threshold identities and inequalities

$$(2.4) \quad \text{lct}(I) = \text{lct}(\bar{I}), \quad \text{lct}(I + J) \leq \text{lct}(I) + \text{lct}(J), \quad \text{lct}(B^N) = \frac{r}{N}$$

when B is generated by r independent linear forms. Restriction to a smooth section cannot increase the threshold. These facts hold for formal ideals as well; see [1, Proposition 2.10] and the valutive treatment in [8]. Restricting to a general l -dimensional section and applying [5] proves $E_l(I) \leq \text{lct}(I)$.

2.2. Valuations and linear bounds. We use $v(0) = +\infty$ and $v(c) = 0$ for $c \in \mathbb{C}^*$. For $t \leq 0$ set $\mathfrak{a}_t(v) = R$. The relation $w \geq v$ means $w(f) \geq v(f)$ for every $f \in R$. Log discrepancy and quasi-monomial valuations are used with the conventions of [8, Sections 3 and 5]. In particular, on a log smooth model, log discrepancy is the weighted sum of the discrepancies of the divisors defining a quasi-monomial valuation.

Lemma 2.1. *If $s = v(\mathfrak{m}) > 0$ and $A = A(v) < \infty$, then*

$$s \text{ord}_o(f) \leq v(f) \leq A \text{ord}_o(f) \quad (0 \neq f \in \mathfrak{m}).$$

Consequently, for $t > 0$,

$$(2.5) \quad \mathfrak{m}^{\lceil t/s \rceil} \subseteq \mathfrak{a}_t(v) \subseteq \mathfrak{m}^{\lceil t/A \rceil}.$$

Proof. The first inequality follows from $f \in \mathfrak{m}^{\text{ord}_o(f)}$. For the other inequality, Skoda's bound gives $\text{lct}(f) \geq 1/\text{ord}_o(f)$, while the valutive formula for the threshold gives $\text{lct}(f) \leq A(v)/v(f)$. Combining the two proves the assertion.

The Skoda bound and its normalization are recalled in [5, Section 1]; the valuative formula is treated in [8]. The ideal inclusions follow by applying the inequalities to each element. $\square$

2.3. Asymptotic mixed multiplicities.

Theorem 2.2 (Cutkosky). *Let $(S, \mathfrak{n})$ be an analytically unramified equicharacteristic local ring of dimension $r > 0$ with perfect residue field. For a graded family $J_\bullet$ of $\mathfrak{n}$ -primary ideals,*

$$\lim_{m \rightarrow \infty} \frac{r! \ell(S/J_m)}{m^r} = \lim_{m \rightarrow \infty} \frac{e(J_m; S)}{m^r}.$$

Both limits exist.

This is [4, Theorems 5.8 and 6.5]. Reduced local rings essentially of finite type over $\mathbb{C}$ satisfy the hypotheses.

Lemma 2.3. *For v as in Lemma 2.1, all $b_j(v)$ exist and*

$$A(v)^{-j} \leq b_j(v) \leq s^{-j}, \quad b_j(v)^2 \leq b_{j-1}(v)b_{j+1}(v) \quad (1 \leq j < n).$$

Moreover $b_n(v) = \text{vol}(v)$, and $w \geq v$ implies $b_j(w) \leq b_j(v)$ whenever both discrepancies are finite.

Proof. For each j , choose $n - j$ sufficiently general linear forms such that, on the resulting regular local quotient S_j of dimension j ,

$$e_j(\mathfrak{a}_m(v)) = e(\mathfrak{a}_m(v)S_j; S_j) \quad \text{for every integer } m \geq 1.$$

Such a choice is possible: the general-section formula for mixed multiplicities imposes a nonempty Zariski open condition for each fixed ideal, and there are only countably many ideals. Over the uncountable field $\mathbb{C}$, these conditions can be satisfied simultaneously. See [7, Chapter 17] for the general-section formula. The images of the valuation ideals form a graded primary family on S_j . Theorem 2.2 proves existence of the required limit. For $j = n$ it identifies that limit with volume.

Inclusions (2.5), monotonicity of mixed multiplicities, and $e_j(\mathfrak{m}^k) = k^j$ give the bounds. The finite-level inequalities $e_j(I)^2 \leq e_{j-1}(I)e_{j+1}(I)$ pass to the limit; see [5, Lemma 2.1]. Finally $w \geq v$ gives $\mathfrak{a}_m(v) \subseteq \mathfrak{a}_m(w)$. Mixed multiplicity decreases under enlargement of an ideal, which proves the last assertion. $\square$

2.4. Thresholds of graded sequences. For a graded sequence $I_\bullet$, write $u(I_\bullet) = \inf_{m \geq 1} u(I_m)/m$. The asymptotic threshold has the descriptions

$$(2.6) \quad \text{lct}(I_\bullet) = \sup_{m \geq 1} m \text{lct}(I_m) = \inf_u \frac{A(u)}{u(I_\bullet)}.$$

Only valuations with positive denominator enter the infimum. For primary sequences with linear bounds they are centered at o . These formulas, and attainment by a valuation, are discussed in [8, Sections 6–7].

Lemma 2.4. *One has $v(\mathbf{a}_\bullet(v)) = 1$. If $w(\mathbf{a}_\bullet(v)) \geq 1$, then $w \geq v$. Furthermore,*

$$(2.7) \quad \text{DP}(v) \leq \text{lct}(\mathbf{a}_\bullet(v)) \leq A(v).$$

Proof. The definition gives $v(\mathbf{a}_m(v)) \geq m$. Fix $f \in \mathfrak{m} \setminus \{0\}$ and let $m_k = \lfloor kv(f) \rfloor$. Since $f^{m_k} \in \mathbf{a}_{m_k}(v)$, the opposite bound follows on letting $k \rightarrow \infty$. If $w(\mathbf{a}_\bullet(v)) \geq 1$, the same containment gives $kw(f) \geq m_k$, hence $w(f) \geq v(f)$.

The Demailly–Pham inequality for $\mathbf{a}_m(v)$ reads

$$m \text{lct}(\mathbf{a}_m(v)) \geq \sum_{j=0}^{n-1} \frac{e_j(\mathbf{a}_m(v))/m^j}{e_{j+1}(\mathbf{a}_m(v))/m^{j+1}}.$$

Its limit, (2.6), and the preceding normalization give (2.7). $\square$

2.5. Numerical monotonicity. Consider the positive log-convex region

$$\mathcal{D} = \{(t_1, \dots, t_n) > 0 : t_j^2 \leq t_{j-1}t_{j+1} \ (1 \leq j < n), \ t_0 = 1\}$$

and $F(t) = \sum_{j=0}^{n-1} t_j/t_{j+1}$. We use the monotonicity in [2, Proposition 10], together with its equality criterion for comparable vectors.

Lemma 2.5. *The region $\mathcal{D}$ is convex and F is nonincreasing along coordinatewise increasing line segments in $\mathcal{D}$. If $q \leq p$ in $\mathcal{D}$, then $F(q) = F(p)$ if and only if $q = p$.*

Proof. For two log-convex sequences, Cauchy–Schwarz shows that their convex combination is log-convex: with $z_j = (1-t)p_j + tq_j$,

$$z_j \leq (1-t)\sqrt{p_{j-1}p_{j+1}} + t\sqrt{q_{j-1}q_{j+1}} \leq \sqrt{z_{j-1}z_{j+1}}.$$

For $1 \leq j < n$, direct differentiation gives

$$\partial_j F = -\frac{t_{j-1}}{t_j^2} + \frac{1}{t_{j+1}} \leq 0, \quad \partial_n F = -\frac{t_{n-1}}{t_n^2} < 0.$$

The derivative signs prove monotonicity along each such segment.

To prove the equality assertion, suppose $q \leq p$ and $F(q) = F(p)$. Along $z(t) = q + t(p - q)$ all derivative contributions are nonpositive, and their sum must vanish identically. If $p \neq q$, let j be the largest index with $p_j > q_j$. For $j = n$ the last derivative makes the sum strictly negative. For $j < n$ vanishing forces $z_j(t)^2 = z_{j-1}(t)z_{j+1}(t)$ for every t . But z_{j+1} is constant, so the right side has degree at most one in t , whereas the coefficient of t^2 on the left is $(p_j - q_j)^2 > 0$. This is impossible. $\square$

3. HYPERSURFACE SECTIONS

The hypersurface identity below relates the volume of a valuation to its value on each function. Applied to a general hyperplane, it identifies the final mixed-multiplicity ratio needed for the induction in Section 4.

Proposition 3.1. *For $0 \neq f \in \mathfrak{m}$ and v of finite discrepancy,*

$$(3.1) \quad \lim_{m \rightarrow \infty} \frac{e(\mathfrak{a}_m(v); R/(f))}{m^{n-1}} = v(f) \operatorname{vol}(v).$$

Proof. First assume that f is irreducible. Put $s = v(f)$, $H(t) = \ell(R/\mathfrak{a}_t(v))$, and $L(t) = \ell(R/(\mathfrak{a}_t(v) + (f)))$. Multiplicativity of v gives the exact colon identity $(\mathfrak{a}_t(v) : f) = \mathfrak{a}_{t-s}(v)$. Thus multiplication by f gives

$$0 \longrightarrow R/\mathfrak{a}_{t-s}(v) \xrightarrow{f} R/\mathfrak{a}_t(v) \longrightarrow R/(\mathfrak{a}_t(v) + (f)) \longrightarrow 0,$$

and consequently $L(t) = H(t) - H(t-s)$. The ring $S = R/(f)$ is reduced and excellent. Theorem 2.2 applied to $\mathfrak{a}_m(v)S$ supplies a number B such that

$$L(m) = \frac{B}{(n-1)!} m^{n-1} + o(m^{n-1}), \quad B = \lim_{m \rightarrow \infty} \frac{e(\mathfrak{a}_m(v); S)}{m^{n-1}}.$$

Both L and H are nondecreasing, so comparison with integer floors and ceilings extends their asymptotic formulas to real arguments. Since $H(0) = 0$, telescoping gives

$$\begin{aligned} H(ks) &= \sum_{i=1}^k L(is) \\ &= \frac{Bs^{n-1}}{(n-1)!} \sum_{i=1}^k i^{n-1} + o(k^n) = \frac{Bs^{n-1}}{n!} k^n + o(k^n). \end{aligned}$$

On the other hand, the definition of volume gives $H(ks) = \operatorname{vol}(v) s^n k^n / n! + o(k^n)$. Hence $B = s \operatorname{vol}(v)$.

For general f , factor $f = c \prod_i f_i^{r_i}$ in the regular local unique factorization domain R , where c is a unit and the f_i are pairwise nonassociate irreducibles. The associativity formula for multiplicity [7, Chapter 11] gives

$$e(\mathfrak{a}_m(v); R/(f)) = \sum_i r_i e(\mathfrak{a}_m(v); R/(f_i)).$$

Indeed the generic length along (f_i) is r_i . Apply the irreducible case and use $v(f) = \sum_i r_i v(f_i)$. $\square$

Corollary 3.2. *One has $b_{n-1}(v) = v(\mathfrak{m}) \operatorname{vol}(v)$.*

Proof. Choose a linear form h which is sufficiently general for the mixed-multiplicity section formula for every $\mathfrak{a}_m(v)$, and such that $v(h) = v(\mathfrak{m})$. The latter condition excludes a proper linear subspace of the space of linear forms. Then $e_{n-1}(\mathfrak{a}_m(v)) = e(\mathfrak{a}_m(v); R/(h))$ for all m . Proposition 3.1 gives the conclusion. $\square$

4. DIVISORIAL RIGIDITY AND EQUALITY IDEALS

The primary classification and the following auxiliary divisorial statement will be proved simultaneously.

Proposition 4.1 (Primary classification and divisorial rigidity).

- (i) *An $\mathfrak{m}$ -primary ideal $I \subset R$ satisfies $\text{DP}(I) = \text{lct}(I)$ if and only if there are regular parameters $z_1, \dots, z_n$ of $\hat{R}$ and positive integers $d_1, \dots, d_n$ such that*

$$(4.1) \quad \overline{I\hat{R}} = \overline{(z_1^{d_1}, \dots, z_n^{d_n})}.$$

The same assertion holds for $\hat{\mathfrak{m}}$ -primary ideals of $\hat{R}$, where $\hat{\mathfrak{m}} = \mathfrak{m}\hat{R}$.

- (ii) *If v is divisorial, centered at o , and $\text{DP}(v) = A(v)$, then its extension to $\hat{R}$ is monomial in regular parameters.*

Throughout this section a divisorial valuation is first normalized as $v = \text{ord}_E$, so its value group is $\mathbb{Z}$. Positive rescaling does not affect the assertion of coordinate monomiality. Geometric constructions are performed on a smooth algebraic representative of the germ. Completion will be used only for the coordinate classification.

4.1. Integral closures and the threshold-computing divisor. We first record a consequence of numerical monotonicity and the multiplicity theorem of Rees. Recall that the latter says that, for nested primary ideals $K \subseteq L$ in a formally equidimensional local ring, $e(K) = e(L)$ implies $\overline{K} = \overline{L}$; see [7, Chapter 11]. All rings to which we apply it are regular.

Lemma 4.2 (The valuation ideals of an equality ideal). *Let I be $\mathfrak{m}$ -primary with $\text{DP}(I) = \text{lct}(I)$, and let ord_F compute $\text{lct}(I)$. Set $q = \text{ord}_F(I)$ and $w = q^{-1} \text{ord}_F$. Then*

$$(4.2) \quad \overline{I^m} = \mathfrak{a}_m(w) \quad (m \geq 1), \quad \text{vol}(w) = e(I), \quad \text{DP}(w) = A(w).$$

In particular, the normalized divisorial valuation computing $\text{lct}(I)$ is unique.

Proof. Since I is $\mathfrak{m}$ -primary, every divisor computing its threshold has center o , and $q > 0$. Blum's results [3, Theorem 1.3 and Proposition 1.5] imply that the valuation-ideal algebra of ord_F is finitely generated: its asymptotic threshold is $A(\text{ord}_F)$, strictly larger than $A(\text{ord}_F) - 1$. Consequently the integer-indexed algebra for $w = \text{ord}_F/q$ is also finitely generated; it is the q -th Veronese of that algebra. There is an integer $M > 0$ such that

$$\mathfrak{a}_{kM}(w) = \mathfrak{a}_M(w)^k \quad (k \geq 1).$$

Here and below, choosing such a Veronese means choosing degrees in which the graded algebra is generated by its first piece.

The normalization gives $w(I) = 1$, whence $I^m \subseteq \mathfrak{a}_m(w)$. Monotonicity of mixed multiplicities therefore gives $b_j(w) \leq e_j(I)$ for every j . Both sequences

are positive and log-convex. Applying Lemma 2.5, and then the DP inequality for valuation ideals, yields

$$(4.3) \quad \text{DP}(I) \leq \text{DP}(w) \leq \text{lct}(\mathfrak{a}_\bullet(w)) \leq A(w) = \text{lct}(I).$$

The two ends coincide by hypothesis. Thus every inequality is an equality. The strict negativity of the last derivative in Lemma 2.5 forces $b_n(w) = e_n(I)$: if the last coordinates differed, integrating that derivative along the segment would make the first inequality strict. Hence $\text{vol}(w) = e(I)$.

For every positive integer k we have

$$I^{kM} \subseteq \mathfrak{a}_{kM}(w), \quad e(I^{kM}) = (kM)^n e(I) = e(\mathfrak{a}_{kM}(w)).$$

The last equality follows from the generated Veronese and the definition of $b_n(w)$. A valuation ideal is integrally closed: an integral equation for an element of smaller value would have its leading power as its unique term of least value, which is impossible. Rees's theorem consequently gives $\overline{I^{kM}} = \mathfrak{a}_{kM}(w)$.

Now fix any $m \geq 1$. The inclusion $\overline{I^m} \subseteq \mathfrak{a}_m(w)$ again follows from integral closedness. Conversely, if $f \in \mathfrak{a}_m(w)$, then $f^M \in \mathfrak{a}_{mM}(w) = \overline{I^{mM}}$. For every divisorial test valuation u , the valuative criterion for integral closure gives $Mu(f) \geq mMu(I)$. Dividing by M and applying the same criterion to I^m shows $f \in \overline{I^m}$. This proves (4.2).

If another divisor computes the threshold, normalize it to obtain w' with $w'(I) = 1$. The argument gives $\mathfrak{a}_m(w') = \overline{I^m} = \mathfrak{a}_m(w)$ for all integers m . These integer thresholds recover every function value, since

$$w(f) = \lim_{k \rightarrow \infty} \frac{1}{k} \max\{m \in \mathbb{Z}_{\geq 0} : f^k \in \mathfrak{a}_m(w)\}.$$

Thus $w' = w$, proving uniqueness. $\square$

Lemma 4.3 (Extraction of the equality divisor). *Let $v = \text{ord}_E$ satisfy $\text{DP}(v) = A(v)$ in dimension $n \geq 2$. Then E is a Kollár component. Moreover, one can choose an integer $N > 0$ such that, with $I = \mathfrak{a}_N(v)$,*

$$(4.4) \quad \mathfrak{a}_{kN}(v) = I^k, \quad v(I) = N, \quad e_j(I) = N^j b_j(v), \quad \text{DP}(I) = \text{lct}(I) = \frac{A(v)}{N}.$$

Proof. The DP-threshold chain gives $\text{lct}(\mathfrak{a}_\bullet(v)) = A(v)$. Blum's finite-generation result applies as in the preceding proof. Choose a generated Veronese, giving the first identity in (4.4). Since $v(\mathfrak{a}_\bullet(v)) = 1$, taking the infimum along this cofinal subsequence gives $v(I)/N = 1$. Mixed multiplicities of powers scale by $e_j(I^k) = k^j e_j(I)$, proving the third identity. Finally, the ordinary DP inequality and evaluation at v give

$$\frac{A(v)}{N} = \text{DP}(I) \leq \text{lct}(I) \leq \frac{A(v)}{v(I)} = \frac{A(v)}{N}.$$

By Lemma 4.2, ord_E is the unique divisorial valuation computing $\text{lct}(I)$, up to scaling. For every divisor F with $\text{ord}_F(\mathfrak{a}_\bullet(v)) > 0$, the generated

Veronese gives

$$\mathrm{ord}_F(\mathfrak{a}_\bullet(v)) = \frac{\mathrm{ord}_F(I)}{N}.$$

If F is different from E as a divisorial valuation, uniqueness therefore implies

$$A(v) < \frac{A(\mathrm{ord}_F)}{\mathrm{ord}_F(\mathfrak{a}_\bullet(v))}.$$

For divisors with zero denominator the right side is interpreted as $+\infty$. Blum's plt-extraction criterion [3, Proposition 4.4] now applies. It extracts E as the only exceptional divisor, with (Y, E) purely log terminal and $-(K_Y + E)$ relatively ample. This is precisely the Kollár-component property, since the center is the closed point and $n \geq 2$. $\square$

4.2. Descent to a general hyperplane.

Lemma 4.4 (Hyperplane descent). *Use the notation of Lemma 4.3. Choose a sufficiently general linear form h , and put*

$$H = (h = 0), \quad S = R/(h), \quad J = IS, \quad s = v(h) = v(\mathfrak{m}).$$

Then

$$(4.5) \quad \mathrm{DP}(J) = \mathrm{lct}_H(J) =: c_H, \quad A(v) = s + Nc_H.$$

Proof. Choose h to have value $s = v(\mathfrak{m})$ and to satisfy the general-section identities $e_j(J) = e_j(I)$ for $0 \leq j \leq n-1$. Here the mixed multiplicities of J are computed with the maximal ideal of the $(n-1)$ -dimensional regular ring S . The bad choices for the first condition form a proper linear subspace, and the section identities hold on a nonempty open set of choices. Corollary 3.2 and (4.4) give the exact ratio

$$(4.6) \quad \frac{e_{n-1}(I)}{e_n(I)} = \frac{b_{n-1}(v)}{Nb_n(v)} = \frac{s}{N}.$$

The pair (H, J^{c_H}) is log canonical near o . Inversion of adjunction for the smooth Cartier divisor H implies that $(X, H + I^{c_H})$ is log canonical near H ; we use [6, Theorem 1.1]. In this notation an ideal boundary means that the discrepancy of u is $A_X(u) - u(h) - c_H u(I)$. Evaluating its nonnegativity at v gives

$$c_H \leq \frac{A(v) - s}{N} = \mathrm{DP}(I) - \frac{e_{n-1}(I)}{e_n(I)} = \mathrm{DP}(J) \leq c_H.$$

The last inequality is the DP inequality on H ; in dimension one it is the elementary equality between reciprocal order and threshold. All inequalities are equalities. This proves both assertions of (4.5). $\square$

4.3. Rees algebras and hyperplane restriction. The next step concerns the scheme structure of the hyperplane in the degeneration, not just its support. Put $\mathfrak{a}_m = \mathfrak{a}_m(v)$ for $m \in \mathbb{Z}$, with $\mathfrak{a}_m = R$ for $m \leq 0$, and define

$$T = \bigoplus_{m \in \mathbb{Z}} \mathfrak{a}_m t^{-m} \subset R[t, t^{-1}], \quad G = T/(t) = \text{gr}_v R,$$

$$\mathfrak{b}_m = (\mathfrak{a}_m + (h))/(h) \subset S, \quad T_H = \bigoplus_{m \in \mathbb{Z}} \mathfrak{b}_m t^{-m} \subset S[t, t^{-1}].$$

These are extended Rees algebras: allowing negative indices puts t in the algebra, and setting $t = 0$ produces the associated graded ring. They are finitely generated by Lemma 4.3 and passage to a quotient. Let $h_0 = \text{in}_v(h) \in G_s$.

Lemma 4.5 (The quotient algebra). *With the preceding notation,*

$$(4.7) \quad T/(ht^{-s}) \simeq T_H, \quad T_H/(t) \simeq G/(h_0), \quad \mathfrak{b}_{kN} = J^k \quad (k \geq 1).$$

Moreover, T_H is a domain and t is a non-zero divisor on T_H .

Proof. For every integer m , the valuation identity gives

$$(4.8) \quad \mathfrak{a}_m \cap hR = h\mathfrak{a}_{m-s}.$$

Indeed, $hg \in \mathfrak{a}_m$ if and only if $s + v(g) \geq m$. The natural graded surjection $T \rightarrow T_H$ has in degree m exactly the kernel in (4.8). This kernel is the degree- m part of the principal ideal (ht^{-s}) . This proves the first isomorphism; setting $t = 0$ proves the second. The identity $\mathfrak{b}_{kN} = J^k$ follows by reducing $\mathfrak{a}_{kN} = I^k$ modulo h . Since T_H is a subring of the domain $S[t, t^{-1}]$, it is a domain. In particular t is a non-zero divisor on it. $\square$

4.4. Normality and reducedness of the hyperplane degeneration.

Write $\mathcal{X} = \text{Spec } T$, $X_0 = \text{Spec } G$, and $\mathcal{H} = V(ht^{-s}) \subset \mathcal{X}$. The divisor $\mathcal{H}$ is Cartier and prime, and after inverting t it is $H \times \mathbb{G}_m$. The preceding domain assertion shows it has no component or embedded part supported over $t = 0$. Thus it is exactly the scheme-theoretic closure of $H \times \mathbb{G}_m$.

Lemma 4.6 (Compatibility with the hyperplane). *The pair $(\mathcal{X}, X_0)$ is purely log terminal, and*

$$(4.9) \quad (\mathcal{X}, X_0 + \mathcal{H}) \text{ is log canonical.}$$

Proof. On a common log resolution of (X, H, I) , write $I\mathcal{O}_Y = \mathcal{O}_Y(-F)$ and include E among the resolved divisors. Choose an integer $r > c_H$ and sufficiently general elements $f_1, \dots, f_r$ of a finite-dimensional generating space for I . Set

$$D = \frac{c_H}{r} \sum_{i=1}^r \text{div}(f_i), \quad \Gamma = H + D.$$

The moving parts of these divisors on Y are base-point-free. Bertini's theorem lets us choose them smooth and transverse to each other and to the fixed resolution divisor. Their coefficients are $c_H/r < 1$, while their fixed part is

exactly $c_H F$. Consequently the log-canonicity inequalities for $(X, H + I^{c_H})$ imply those for (X, Γ) . Also $v(f_i) = N$ for general f_i , so

$$A_{X, \Gamma}(v) = A(v) - s - c_H N = 0.$$

After shrinking the smooth germ, K_X is trivial and all the divisors just chosen are principal. Hence $K_X + \Gamma \sim_{\mathbb{Q}} 0$ locally, (X, Γ) is log canonical, and o is an lc center. Thus Γ is a local $\mathbb{Q}$ -complement with E as an lc place.

The Kollár-component degeneration theorem [12, Lemma 2.17] now gives

$$(4.10) \quad (\mathcal{X}, X_0) \text{ is plt}, \quad (\mathcal{X}, X_0 + \Gamma_{\mathcal{X}}) \text{ is lc},$$

where $\Gamma_{\mathcal{X}}$ is the closure of the boundary on the nonzero fibers. Its coefficient along $\mathcal{H}$ is one. Removing the other effective boundary components yields (4.9). $\square$

Proposition 4.7 (Normal quotient and reduced central fiber). *The domain T_H is normal, and $G/(h_0)$ is reduced.*

Proof. For an exceptional divisor F' over $\mathcal{X}$ whose center is contained in X_0 , removing $X_0 = \text{div}(t)$ from (4.9) gives

$$A_{\mathcal{X}, \mathcal{H}}(F') = A_{\mathcal{X}, X_0 + \mathcal{H}}(F') + \text{ord}_{F'}(t) > 0.$$

If the center is not contained in X_0 , its generic point lies in $\mathcal{X} \setminus X_0 = X \times \mathbb{G}_m$. There the pair is the product of the smooth pair (X, H) with $\mathbb{G}_m$, and its exceptional discrepancies are positive. Thus $(\mathcal{X}, \mathcal{H})$ is plt. A coefficient-one prime divisor in a plt pair is normal, so T_H is normal. This standard normality statement and adjunction for plt pairs are reviewed in [9, Chapter 5].

Set $C = G/(h_0)$. Lemma 4.6 implies that X_0 is normal and klt by adjunction, hence G is Cohen–Macaulay [9, Chapter 5]. Here X_0 is Cartier; its different is zero at every codimension-one point, as can also be seen from the following local calculation. Let η be a codimension-one point of X_0 at which h_0 vanishes. The ring $\mathcal{O}_{X_0, \eta}$ is a discrete valuation ring. The ambient local ring $B = \mathcal{O}_{\mathcal{X}, \eta}$ has dimension two, and $B/(t) = \mathcal{O}_{X_0, \eta}$. Lifting a generator of the maximal ideal of this discrete valuation ring shows that the maximal ideal of B is generated by two elements, one of them t . Thus B is regular.

Ordinary adjunction for the lc pair (4.9), at this regular point of the ambient space, says that the coefficient of $\mathcal{H}|_{X_0}$ at η is at most one. That coefficient is precisely $\text{ord}_{\eta}(h_0)$, a positive integer. It is therefore one. Every irreducible component of C consequently has multiplicity one. Finally, h_0 is a non-zero divisor in the domain G , so C is Cohen–Macaulay and has no embedded associated primes. A Noetherian ring with no embedded associated primes which is reduced at its minimal primes is reduced. This proves the claim. $\square$

4.5. Simultaneous induction and lifting of parameters. We now prove the two statements of Proposition 4.1 together. In dimension one a centered divisorial valuation is a multiple of ordinary order, and every primary ideal is

a power of the maximal ideal. Both assertions follow immediately. Suppose the ideal assertion is known in dimension $n - 1$, and let $v = \text{ord}_E$ be an equality valuation in dimension n .

Use the notation and results of Lemmas 4.3–4.6 and Proposition 4.7. We pass to the completions of R and S at this point. This passage preserves every finite-colength ideal quotient and its associated graded piece. It also preserves the normality of T_H : the completion map of the excellent regular local ring S is a regular morphism, and its base change to the finite-type normal S -algebra T_H is normal; see [11, Tag 0BFK]. The base-changed algebra embeds in $\widehat{S}[t, t^{-1}]$ by flatness. We continue to write T_H , $\mathfrak{b}_m$, and J for these extensions.

The finite-discrepancy valuation itself extends uniquely to $\widehat{R}$. If $f \neq 0$ has order k , its sufficiently long polynomial truncations have order k and values at most $A(v)k$, whereas their successive differences have arbitrarily large value by Lemma 2.1. Their values therefore stabilize. This defines the extension, and the same argument preserves products, sums, valuation ideals, and initial forms.

Lemma 4.8 (Identification of the quotient filtration). *Under the dimension- $(n - 1)$ ideal assertion, there are formal regular parameters $y_1, \dots, y_{n-1}$ of $\widehat{S}$ and positive integers $a_1, \dots, a_{n-1}$ such that*

$$(4.11) \quad \mathfrak{b}_m = \{g : w(g) \geq m\} \quad (m \geq 0), \quad w(y_i) = a_i,$$

where w is the indicated monomial valuation. In particular, $G/(h_0) \simeq \mathbb{C}[Y_1, \dots, Y_{n-1}]$, with $\deg Y_i = a_i$.

Proof. First every power $J^k = \mathfrak{b}_{kN}$ is integrally closed. Indeed, if g is integral over J^k , an integral equation for g , multiplied by the corresponding power of t^{-kN} , shows that gt^{-kN} is integral over T_H . It lies in the fraction field of T_H , since $\widehat{S}$ and t lie in T_H . Normality gives $gt^{-kN} \in T_H$, hence $g \in J^k$.

By Lemma 4.4, J is an equality ideal. The induction hypothesis and integral closedness give formal regular parameters y_i and positive integers d_i such that

$$J = \overline{(y_1^{d_1}, \dots, y_{n-1}^{d_{n-1}})}.$$

Let w be the rational monomial valuation with $w(y_i) = N/d_i$. The Newton polyhedron of this pure-power ideal is $\{u \geq 0 : \sum_i u_i/d_i \geq 1\}$. Integral closure of its k -th power is therefore $\{g : w(g) \geq kN\}$. Since J^k is integrally closed and has the same integral closure as that power, we obtain

$$(4.12) \quad J^k = \{g : w(g) \geq kN\} \quad (k \geq 1).$$

We must identify the intermediate filtration levels, not just this Veronese. For $m \geq 0$ and $g \in \widehat{S}$,

$$(4.13) \quad g \in \mathfrak{b}_m \iff g^N \in \mathfrak{b}_{mN} = J^m.$$

The forward direction uses multiplicativity of the filtration. For the reverse, $(gt^{-m})^N \in T_H$, so gt^{-m} is integral over the normal domain T_H and belongs to it. Combining (4.13) with (4.12) yields $\mathfrak{b}_m = \{g : w(g) \geq m\}$ for every m .

The numbers N/d_i are in fact integers. If $r = w(y_i)$ were not integral, set $k = \lfloor r \rfloor$. The class of y_i in $\mathfrak{b}_k/\mathfrak{b}_{k+1}$ is nonzero. Choose an integer $p > 0$ with $p(r - k) \geq 1$. Then $y_i^p \in \mathfrak{b}_{pk+1}$, so this nonzero class is nilpotent in $G/(h_0)$. This contradicts its reducedness from Proposition 4.7. Thus $a_i = N/d_i$ are positive integers. The integer-indexed graded ring of this monomial valuation is the asserted weighted polynomial ring. $\square$

Divisorial induction step. Let $y_1, \dots, y_{n-1}$ be the parameters in Lemma 4.8. Choose a lift $z_i \in \mathfrak{a}_{a_i}(v)\widehat{R}$ of each y_i . Its value is exactly a_i , since a higher value would put y_i in $\mathfrak{b}_{a_i+1}$. The elements

$$z_1, \dots, z_{n-1}, h$$

form a regular system of parameters of $\widehat{R}$: their images in $\widehat{R}/(h)$ are regular parameters, and h has ordinary order one.

Their initial forms generate G . Indeed, take a homogeneous element $g \in G$ of degree d . Modulo h_0 , express it as a weighted homogeneous polynomial in the images of the $\text{in}_v(z_i)$. Subtract that polynomial from g . The remainder is $h_0 g_1$ for a homogeneous element g_1 of degree $d - s$. Since $s > 0$, induction on d expresses g_1 , and then g , in the proposed generators. The induction starts in degree zero, where $G_0 = \mathbb{C}$.

We obtain a graded surjection

$$\mathbb{C}[Z_1, \dots, Z_{n-1}, U] \longrightarrow G, \quad \deg Z_i = a_i, \quad \deg U = s.$$

The target is a domain of dimension n . For example, its dimension follows from the degree- n cumulative Hilbert growth $\ell(R/\mathfrak{a}_m(v)) \sim \text{vol}(v)m^n/n!$ and finite generation. The kernel is a prime ideal of height zero in a polynomial ring, and so is zero. Thus no nonzero polynomial in these initial forms can cancel. For any formal series in these parameters, there are only finitely many monomials of bounded weight, and its least-weight part has nonzero initial form. Its value is therefore the minimum of the weights of its monomials. This is exactly coordinate monomiality of v . $\square$

Ideal induction step. Let K be an $\mathfrak{m}$ -primary equality ideal and choose a divisor $v = \text{ord}_E$ computing its threshold. Put $q = v(K)$. Lemma 4.2 gives $\text{DP}(v) = A(v)$ and

$$(4.14) \quad \overline{K} = \mathfrak{a}_q(v), \quad e(K) = q^n \text{vol}(v).$$

The divisorial assertion just proved gives formal regular parameters z_i with positive integral weights $\alpha_i = v(z_i)$. Hence $\text{vol}(v) = 1/(\alpha_1 \cdots \alpha_n)$. In these parameters $\overline{K}$ is the monomial ideal generated by the monomials with $\sum_i \alpha_i u_i \geq q$.

Let P be its Newton polyhedron and let $Q = \{u \in \mathbb{R}_{\geq 0}^n : \sum_i \alpha_i u_i \geq q\}$. Then $P \subseteq Q$. The multiplicity formula for a primary monomial ideal, together

with (4.14), gives

$$n! \operatorname{Vol}(\mathbb{R}_{\geq 0}^n \setminus P) = \frac{q^n}{\alpha_1 \cdots \alpha_n} = n! \operatorname{Vol}(\mathbb{R}_{\geq 0}^n \setminus Q).$$

The Newton-polyhedron description of integral closure and this multiplicity formula are recalled in [2, 7]. The closed convex polyhedra P and Q must coincide. Indeed, if an axial vertex $(q/\alpha_i)e_i$ of Q were outside P , closedness would give a small neighborhood of it disjoint from P whose intersection with Q has positive volume, contradicting the displayed equality. If all axial vertices belong to P , convexity and stability under addition of the positive orthant give $Q \subseteq P$ directly.

Every vertex of the Newton polyhedron of a monomial ideal is an integral exponent vector of a monomial generator. Thus $d_i = q/\alpha_i$ is a positive integer for every i . The identity $P = Q$ proves $\overline{K} = \overline{(z_1^{d_1}, \dots, z_n^{d_n})}$ in $\hat{R}$.

Conversely, order the positive integers so that $d_1 \leq \dots \leq d_n$. The pure-power ideal has

$$e_j = d_1 \cdots d_j \quad (1 \leq j \leq n), \quad \operatorname{lct} = \sum_{i=1}^n \frac{1}{d_i}.$$

These are the usual mixed-multiplicity and threshold formulas for a diagonal ideal [2]. Hence $\operatorname{DP} = \operatorname{lct}$. Integral closure, formal changes of regular parameters, and completion preserve the invariants in this equality, giving the converse for K .

Finally, an $\hat{\mathfrak{m}}$ -primary ideal of $\hat{R}$ contains $\hat{\mathfrak{m}}^L$ for some L . Truncating a finite generating set modulo $\hat{\mathfrak{m}}^L$ and adjoining the degree- L monomials expresses it as the extension of an $\mathfrak{m}$ -primary ideal in R . Its mixed multiplicities and threshold agree with those of this algebraic ideal (a log resolution remains a log resolution after completion). The assertion therefore holds for formal ideals as well. This completes the simultaneous induction and the proof of Proposition 4.1. $\square$

5. EQUALITY IDEALS IN ARBITRARY CODIMENSION

In this section $S = \mathbb{C}[[x_1, \dots, x_n]]$ and $\mathfrak{n}$ is its maximal ideal. We prove Theorem 1.1. The reduction to primary ideals is separated from the primary classification itself: the numerical and formal approximation lemmas below do not use Section 4.

5.1. Transverse primary truncations.

Lemma 5.1 (Mixed multiplicities of a transverse truncation). *Let $0 \neq I \subsetneq S$ have height at least $l < n$, set $s_j = \sigma_j(I)$ for $0 \leq j \leq l$, and put $r = n - l$. Choose $f_1, \dots, f_l \in I$ and independent linear forms $z_1, \dots, z_r$ as in (2.3). Write $B = (z_1, \dots, z_r)$ and $K_N = I + B^N$. Then K_N is primary to $\mathfrak{n}$ and*

$$(5.1) \quad e_j(K_N) \leq p_j(N), \quad p_j(N) = \begin{cases} s_j, & 0 \leq j \leq l, \\ s_l N^{j-l}, & l < j \leq n. \end{cases}$$

For every integer $N \geq s_l/s_{l-1}$,

$$(5.2) \quad E_n(K_N) \geq E_l(I) + \frac{r}{N}.$$

Proof. The radical of $I + B^N$ is the radical of $I + B$, which is $\mathfrak{n}$ by (2.3). For $j \leq l$, monotonicity of the extended mixed multiplicity gives $e_j(K_N) \leq \sigma_j(I) = s_j$.

For $j = l + q$, where $0 \leq q \leq r$, consider the parameter tuple

$$f_1, \dots, f_l, z_1^N, \dots, z_q^N, z_{q+1}, \dots, z_r.$$

Its first j elements belong to K_N and its last $n - j$ elements belong to $\mathfrak{n}$. The mixed-multiplicity comparison recalled in Section 2 therefore gives

$$\begin{aligned} e_j(K_N) &\leq \ell(S/(f_1, \dots, f_l, z_1^N, \dots, z_q^N, z_{q+1}, \dots, z_r)) \\ &= N^q \ell(S/(f_1, \dots, f_l, z_1, \dots, z_r)) = N^q s_l. \end{aligned}$$

For clarity, this comparison does not require the displayed tuple to be general for K_N . Apply monotonicity to the principal ideals generated by its entries. Their extended mixed multiplicity is the colength of the parameter ideal, because S is regular. This proves (5.1).

The ratios s_j/s_{j-1} are nondecreasing by log convexity. Appending the constant ratio $N \geq s_l/s_{l-1}$ thus makes $p(N)$ a positive log-convex vector. The vector of mixed multiplicities of K_N is also positive and log-convex, and is coordinatewise bounded by $p(N)$. Lemma 2.5 gives

$$E_n(K_N) \geq F(p(N)) = \sum_{j=0}^{l-1} \frac{s_j}{s_{j+1}} + \frac{r}{N}.$$

□

Proposition 5.2 (Persistence of equality under truncation). *In the notation of Lemma 5.1, assume $\text{lct}(I) = E_l(I)$. Then $\text{ht } I = l$ and, for every integer $N \geq s_l/s_{l-1}$,*

$$(5.3) \quad \text{lct}(K_N) = E_n(K_N) = E_l(I) + \frac{r}{N}, \quad e_j(K_N) = p_j(N) \quad (0 \leq j \leq n).$$

Proof. If $\text{ht } I \geq l + 1$, the general-section DP inequality gives $\text{lct}(I) \geq E_{l+1}(I) > E_l(I)$, a contradiction. By (2.4),

$$\text{lct}(K_N) \leq \text{lct}(I) + \text{lct}(B^N) = E_l(I) + \frac{r}{N}.$$

Combine this with Lemma 5.1 and the primary DP inequality to obtain

$$E_l(I) + \frac{r}{N} \leq E_n(K_N) \leq \text{lct}(K_N) \leq E_l(I) + \frac{r}{N}.$$

All terms coincide. The equality assertion in Lemma 2.5, applied to the comparable vectors $(e_j(K_N))$ and $p(N)$, gives equality of every coordinate. □

5.2. Integral closure and the adic limit.

Lemma 5.3 (Continuity of integral closure). *For any proper ideals $I, B \subset S$, with $I \neq 0$, one has*

$$(5.4) \quad \bigcap_{N \geq 1} \overline{I + B^N} = \bar{I}.$$

Moreover, for every $k \geq 1$ there is $N(k)$ such that

$$(5.5) \quad \overline{I + B^N} + \mathfrak{n}^k = \bar{I} + \mathfrak{n}^k \quad (N \geq N(k)).$$

Proof. One inclusion in (5.4) is immediate. For the other, suppose $f \in \overline{I + B^N}$ for every N . The Briançon–Skoda theorem in a regular local ring of dimension n gives, for all $m, N \geq 1$,

$$f^{m+n-1} \in \overline{(I + B^N)^{m+n-1}} \subseteq (I + B^N)^m \subseteq I^m + \mathfrak{n}^N.$$

See [7, Chapter 13]. Krull intersection, for fixed m , gives $f^{m+n-1} \in I^m$. If u is any Rees valuation of I , then $(m+n-1)u(f) \geq mu(I)$. Dividing by m and letting $m \rightarrow \infty$ gives $u(f) \geq u(I)$. The Rees-valuation criterion for integral closure proves $f \in \bar{I}$.

We include the completeness argument needed for (5.5). In a complete Noetherian local ring $(A, \mathfrak{p})$ with residue field $\mathbb{C}$, a decreasing sequence of ideals L_N with intersection zero is eventually contained in each $\mathfrak{p}^k$. Indeed, the images of L_N in $A/\mathfrak{p}^k$ stabilize, since this is a finite-dimensional vector space. Call the stable image V_k . The maps $V_{k+1} \rightarrow V_k$ are surjective: choose N large enough for both images to have stabilized and lift using an element of L_N . If some V_k were nonzero, these surjections would give a compatible sequence with a nonzero k -th component. Completeness would yield a nonzero element of A lying in $L_N + \mathfrak{p}^j$ for all N, j . Every ideal is adically closed by Krull intersection, so this element would belong to every L_N , a contradiction. This proves the assertion, usually called Chevalley’s lemma. Apply it to $A = S/\bar{I}$ and $L_N = \overline{I + B^N}/\bar{I}$, using (5.4). The result is (5.5). $\square$

Corollary 5.4 (A complete-intersection reduction). *Under Proposition 5.2, the ideal $J = (f_1, \dots, f_l) \subset I$ satisfies $\bar{J} = \bar{I}$. In particular, the analytic spread of I equals its height l .*

Proof. The parameter ideal $Q_N = J + (z_1^N, \dots, z_r^N)$ is contained in K_N and has multiplicity $N^r s_l$. Proposition 5.2 gives the same multiplicity for K_N . Rees’s multiplicity theorem yields $\overline{Q_N} = \overline{K_N}$ for all large N . Consequently $I \subseteq \overline{J + B^N}$ for all such N . Lemma 5.3 gives $I \subseteq \bar{J}$, and $J \subset I$ gives the reverse inclusion of integral closures. The analytic spread is at least the height and at most the number l of generators of this reduction; see [7, Proposition 8.3.7]. $\square$

5.3. Compatible coordinate changes. The following lemma concerns equality of ideals, not merely equality of their radicals. We give the complex algebraic jet-group argument; compare the finite-order equivalence theorem and its proof in [13, Section 5].

Lemma 5.5 (From finite jets to formal equivalence). *Let $U, V \subset S$ be ideals. Suppose that, for every $k \geq 2$, there is a $\mathbb{C}$ -algebra automorphism ϕ_k of $S/\mathfrak{n}^k$ carrying the image of U onto the image of V . Then there is a formal $\mathbb{C}$ -algebra automorphism ϕ of S with $\phi(U) = V$.*

Proof. Put $G_k = \text{Aut}_{\mathbb{C}}(S/\mathfrak{n}^k)$. These are algebraic groups: coefficients of the images of the variables give coordinates, with invertible linear part. Let H_k stabilize the image of U , and let T_k carry that image onto the image of V . Then H_k is a closed algebraic subgroup and T_k is a nonempty coset of H_k .

For $m \geq k$, truncation maps G_m to G_k and H_m into H_k . The image of H_m is closed by the homomorphism theorem for algebraic groups [10, Theorem 1.72]. Thus the images of T_m in G_k are nonempty closed cosets. They decrease with m and stabilize by Noetherianity. Denote the stable image by T_k^∞ . Choosing m large enough for stabilization at both levels shows

$$T_{k+1}^\infty \longrightarrow T_k^\infty \quad \text{is surjective.}$$

Choose compatible elements of these sets. Their inverse limit defines an automorphism ϕ of S . For every k it carries $U + \mathfrak{n}^k$ onto $V + \mathfrak{n}^k$. Krull intersection, applied to both inclusions, gives $\phi(U) = V$. $\square$

Lemma 5.6 (Jets of the diagonal models). *Let $d_1, \dots, d_l$ be positive integers, $l < n$, and set*

$$D = \overline{(x_1^{d_1}, \dots, x_l^{d_l})}, \quad D_N = \overline{(x_1^{d_1}, \dots, x_l^{d_l}, x_{l+1}^N, \dots, x_n^N)}.$$

If L is the least common multiple of the d_i , then

$$(5.6) \quad D \subseteq D_N \subseteq D + \mathfrak{n}^{\lceil N/L \rceil}.$$

Proof. Integral closures of monomial ideals are described by their Newton polyhedra [7, Chapter 1]. Membership of a monomial x^α in D_N means

$$\sum_{i=1}^l \frac{\alpha_i}{d_i} + \frac{1}{N} \sum_{i=l+1}^n \alpha_i \geq 1.$$

If this monomial does not belong to D , its first sum is strictly less than one. That sum is an integral multiple of $1/L$, so it is at most $1 - 1/L$. Hence $\sum_{i=l+1}^n \alpha_i \geq N/L$ and the monomial belongs to $\mathfrak{n}^{\lceil N/L \rceil}$. Applying this to the monomial generators proves the second inclusion. The first follows from the defining containment of pure-power ideals. $\square$

5.4. Reduction to the primary classification.

Theorem 5.7 (Primary-to-general reduction). *Fix n . Suppose every $\mathfrak{n}$ -primary ideal $K \subset S$ with $\text{lct}(K) = E_n(K)$ has integral closure equal to that of a pure-power ideal in some regular parameters of S . Then Theorem 1.1 holds in dimension n , for every $1 \leq l \leq n$.*

Proof. The forward implication for $l = n$ is the hypothesis. Suppose $l < n$ and $\text{lct}(I) = E_l(I)$. Choose B as in Lemma 5.1. Proposition 5.2 makes $K_N = I + B^N$ a primary equality ideal for every sufficiently large integer N .

The primary hypothesis supplies a formal automorphism Φ_N and ordered positive integers $a_{1,N} \leq \dots \leq a_{n,N}$ such that $\overline{K_N}$ is the image of the integral closure of the corresponding pure-power ideal. For such an ideal, $e_j = a_{1,N} \cdots a_{j,N}$; see [2]. Mixed multiplicities are preserved by integral closure and automorphisms, since automorphisms preserve $\mathfrak{n}$. Equation (5.3) therefore forces

$$(5.7) \quad a_{j,N} = d_j := \frac{s_j}{s_{j-1}} \quad (1 \leq j \leq l), \quad a_{j,N} = N \quad (l < j \leq n).$$

In particular the d_j are positive integers, independent of N . They are ordered by log convexity. With the notation of Lemma 5.6, we have

$$(5.8) \quad \overline{K_N} = \Phi_N(D_N).$$

Fix $k \geq 2$. Choose N large enough for (5.5) and for $\lceil N/L \rceil \geq k$. Lemma 5.6 and the preservation of $\mathfrak{n}^k$ under Φ_N give

$$\begin{aligned} \Phi_N(D) + \mathfrak{n}^k &= \Phi_N(D_N) + \mathfrak{n}^k \\ &= \overline{K_N} + \mathfrak{n}^k = \overline{I} + \mathfrak{n}^k. \end{aligned}$$

Thus D and $\overline{I}$ are equivalent at every finite order. The changes Φ_N need not themselves form a compatible sequence. Lemma 5.5 replaces their finite-order existence by a compatible sequence and produces a formal automorphism Φ with $\Phi(D) = \overline{I}$. Setting $y_i = \Phi(x_i)$ proves (1.2), and (5.7) gives the stated exponents.

Conversely, suppose (1.2) holds with $d_1 \leq \dots \leq d_l$. The diagonal formulas give

$$\sigma_j(I) = d_1 \cdots d_j \quad (1 \leq j \leq l), \quad \text{lct}(I) = \sum_{j=1}^l \frac{1}{d_j}.$$

For the multiplicity formula, restrict to a general l -dimensional section: the images of $y_1, \dots, y_l$ are regular parameters, and integral dependence is preserved under the quotient map. Apply the primary diagonal formula on that section and its further general sections. The threshold formula follows from the Newton polyhedron, or from the monomial valuation with weights $1/d_1, \dots, 1/d_l$ and zero weights in the remaining directions. These invariants are unchanged by integral closure. Substitution gives $E_l(I) = \text{lct}(I)$, proving the converse. $\square$

Proof of Theorem 1.1. For $n = 1$ the assertion is the classification of nonzero proper ideals of a discrete valuation ring. For $n \geq 2$, Proposition 4.1(i) verifies the primary hypothesis of Theorem 5.7, including arbitrary formal primary ideals. Apply that reduction. $\square$

DECLARATION ON THE USE OF ARTIFICIAL INTELLIGENCE

The author selected the research problems. The arguments and exposition were developed with OpenAI Codex and remain subject to the author's mathematical verification.

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